ECE 313 Spring 2026 Project Task 0 & 1

Team1

dhruvv4, vmagow2, brianmk3

Dhruv Vohra, Vedaant Magow, Brian Keating

Task 0

- Q2a: What is the total number of XID errors observed on March 10th, 2024
  - Total XID errors on March 10, 2024: 87

- Q2b: What is the list of unique XID errors on that day?
  - Unique XID list: [74, 43, 45, 13, 119, 31]

Task 1

- Q1: List the first ten data rows (excluding the header row) of the Pandas Dataframe sorted by increasing timestamp.

	Date	datetime	timestamp	node_name	device_id	XID	Tag
0	2022-07-17	2022-07-17T10:27:27.422837-05:00	1.658072e+09	gpua088	0000:c7:00	119	119_GSP_RPC_timeout
1	2022-07-17	2022-07-17T10:29:29.304632-05:00	1.658072e+09	gpua048	0000:07:00	45	045_GPU_PREEMPTIVE_CLEANUP
2	2022-07-17	2022-07-17T11:03:11.093546-05:00	1.658074e+09	gpua001	0000:85:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION
3	2022-07-17	2022-07-17T11:04:15.807907-05:00	1.658074e+09	gpua044	0000:85:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION
4	2022-07-17	2022-07-17T11:10:53.818266-05:00	1.658074e+09	gpua045	0000:07:00	43	043_GPU_STOPPED_PROCESSING
5	2022-07-17	2022-07-17T11:16:30.534940-05:00	1.658075e+09	gpua001	0000:c7:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION
6	2022-07-17	2022-07-17T11:27:40.849346-05:00	1.658075e+09	gpua001	0000:c7:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION
7	2022-07-17	2022-07-17T12:10:20.898006-05:00	1.658078e+09	gpua072	0000:85:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION
8	2022-07-17	2022-07-17T13:21:30.008143-05:00	1.658082e+09	gpua081	0000:85:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION
9	2022-07-17	2022-07-17T13:21:34.279469-05:00	1.658082e+09	gpua005	0000:c7:00	13	013_GPU_GRAPHIC_ENGINE_EXCEPTION

Task 1

- Q1: What is the first timestamp, in both UNIX and human readable datetime, of the XID error observed in the given dataset?

  first time stamp in Unix:

  1658071647.4228365

  first time stamp in human readable format:

  2022-07-17 15:27:27.422836542

- Q2a: What is the probability that a randomly sampled XID from our dataset is XID 43?

  P(XID = 43) = 0.11687569714612182

Task 1

- Q2b: What is the probability that the sampled entry is XID 119?

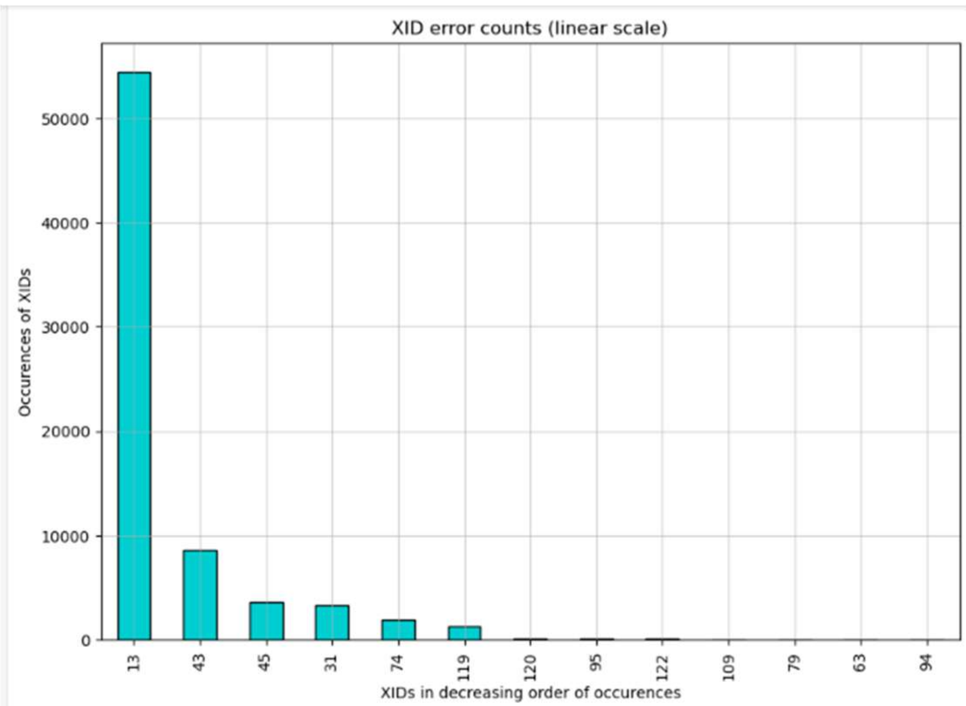
  P(XID = 119) = 0.017724515058356233

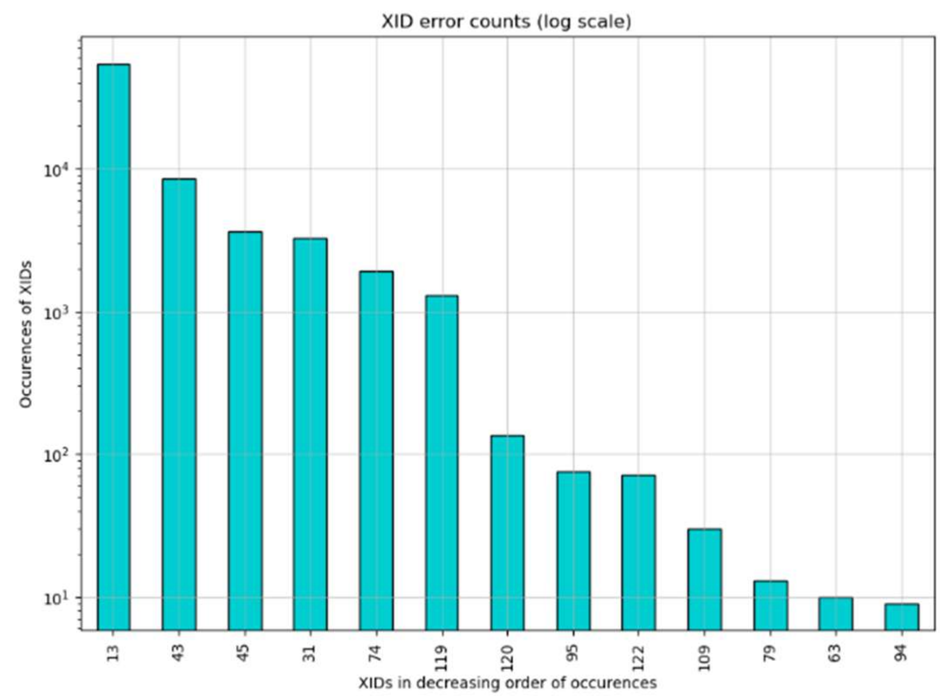
- Q2c: Is the system more likely to encounter user errors that cause XID 43 or hardware errors that cause XID 119?

The system is much more likely to encounter user errors that cause XID 43 than hardware errors that cause XID 119 (0.1168 > 0.0177)

Task 1

- Q3: Show the bar plots below.





Task 1

- Q4a: What is the dominant XID error in our system?

  XID 13

- Q4b: What are the top 5 XIDs that occurred in our dataset?

  XID 13, XID 43, XID 45, XID 31, XID 74
  (ordered from most to least common)

Task 1

- Q4c: What is the fifth most frequent XID a hardware, memory, or interconnect error? What is the tag of this XID? What is the description of this XID?

Fifth most frequent XID (XID 74):

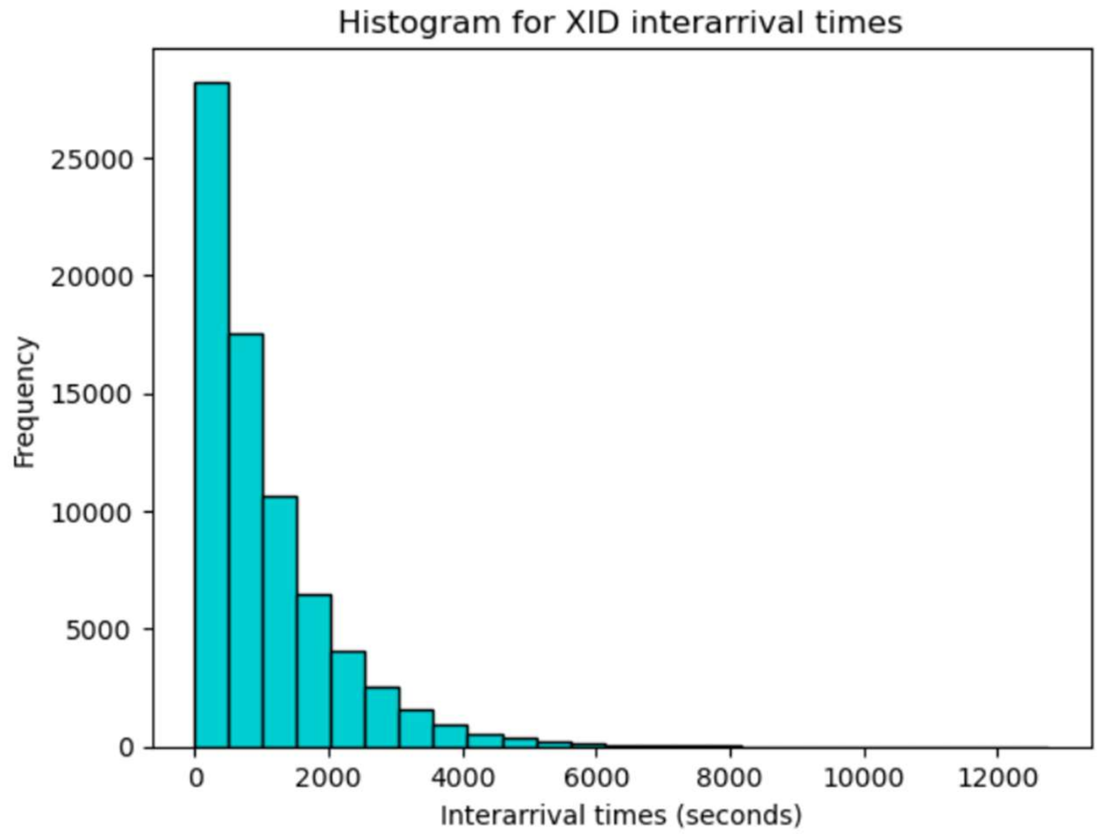
Interconnect Error

Tag = 074_NVLINK_error

Description: NVLink error indicating connection issues between GPUs via NVLink interconnection.

Task 1

- Q5a: Show the histogram plot below.



Task 1

- Q5b: Mean and standard deviation of the interarrival time?

  Mean: 1054.4875208708004 seconds

  Standard Deviation: 1061.6875418103216 seconds

- Q5c: what is the average time in seconds that the user can run his/her job before it is abruptly interrupted by an XID?

  This average time is equal to the mean interarrival time of XIDs: 1054.4875208708004 seconds.

Task 1

- Q6: What is the best fit distribution among these classes of distributions? Report both the fitting statistics (p-value, KS-statistics) outputs and the best fit parameters.

Distribution	KS Statistic	P-value
Exponential	0.0027587078476604687	0.4311377245508982
Normal	0.160304403757268	0.006622516556291391
Uniform	0.7107835116555101	0.006622516556291391
Lognormal	0.03545989968959473	0.9337748344370861

The Exponential distribution is the best fit since its p-value > 0.05 and it has the smallest KS Statistic.
Best Fit Parameters:

Location(start~0) = 0.011411905288696289

Scale(mean) = 1054.4761089655117

Task 1

- Q7a: What is the name of the distribution?

Poisson Distribution

- Q7b: What is the parameter (or parameters) of this distribution that you fitted?

Lambda = 3.4140176049429525 (average errors per hour)

Task 1

- Q7c: Can the system meet this user's requirement? Answer Yes or No and justify.

Yes, Using the Poisson CDF with the lambda parameter determined for this distribution, we found:

P(no more than 5 errors within one hour) = 0.8687652653638034

Therefore, this system meets the user's requirement as this probability is greater than 0.8.